SD-301 — technical document (More Than A Hack 2027, Phase 1)

Team **SD-301**. Output: `outputs/SD-301.csv` (199,319 rows, `frame,label`). Regenerate with `python run_sd301.py`. Everything is tracking-based; **no video decoding, no external dataset, no training on the ground truth.**

1. Approach

Possession is inferred from the tactical Automatic Tracking Data (ATD) with a small, fully explainable model: per-frame evidence, then a sequence decoder that encodes football rules.

1. **Ball clean-up.** Ball gaps up to 0.6 s are bridged linearly; longer gaps stay unknown (the ball is missing on 41 % of graded frames, so unknown must be a first-class value).
2. **Team evidence (ball visible).** Each player gets a weight $\exp(-d^2/2\sigma^2)$ (σ = 2.5 m, ignored beyond 6 m); the evidence is the signed, normalised weight of home minus away, scaled by how close the closest player is. This is a soft nearest-player rule that does not flip in crowds.
3. **Dead-ball evidence (log-odds per frame).** Ball outside the lines (+), ball stationary over 2 s (+), ball travelling > 5 m in 2 s (−), unseen for > 4 s (+, weak), gap that starts or ends within 3 m of a touchline (+), players walking (+, weak).
4. **Evidence when the ball is not tracked (self-supervised).**
5. A *dead-ball* logistic model on collective player movement (mean speed, share running / standing, centroid drift, block length / width over ±1 s).
6. A *possession* logistic model per half on the shape of the two teams (who is further up the pitch, line heights, block length / width, mean speed difference). Features are computed over ±3 s windows (chosen over ±1 s and ±6 s on the *held-out pseudo-label* AUC, never on the possession ground truth). Targets are **pseudo-labels computed from the tracking itself** on frames where the ball *is* tracked (outside lines or stationary ⇒ dead; fast inside the pitch ⇒ alive; clear single-side control ⇒ that side). The models are refitted from scratch in every run, in < 1 s. Held-out AUC on the pseudo-labels: ≈ 0.89 (dead ball, cross-half), ≈ 0.85 (possession).
7. **Restart prior.** Throw-ins, goal kicks, corners and free kicks are awarded against the last side to touch the ball, so after a dead-ball stretch resuming with the *same* side costs ln(0.85/0.15) ≈ 1.7 nats. Implemented with two dead states (dead-after-H, dead-after-A).
8. **Smart Tagging.**
9. `SET PIECES` whose `Team` tag names a side add a soft team bias (0.3 per frame) over the 12 s after the instance starts. `N/A` instances (all throw-ins, all kick-offs) are ignored.
10. Phases of play (build-up, progression, final third, …, and the defending/transition counterparts) name who holds the ball (attacking codes: the tagged side; defending codes: the other side). They last tens of seconds and span several possessions (`DATA.md`), so they are **fallback evidence only**: weight 0.25 on frames where the ball is not tracked, never on frames where measured control exists.
11. **Decoding.** One Viterbi pass per half over the four states with switching costs expressing football: an H↔A hand-over needs ≈ 1.5 s of sustained opposing control, entering or leaving a stoppage ≈ 1 s. A missing ball gives no evidence, so the previous state persists (this is the behaviour the scorer's abstention fill would impose anyway). We never output `X` on graded frames; `X` is written outside the two published halves.

2. Assets used

asset	used	how
tactical ATD (players + ball)	yes	everything above
tactical Smart Tagging	yes	set-piece team bias only
tactical homography	no	tracking is already in pitch coordinates
tactical video	no	never decoded; not needed for this approach
broadcast video / ATD / homography	no	not frame-aligned; dropped to keep the pipeline simple
public 5-minute ground truth	read-only sanity check	see §3

Also used: the published half boundaries (video frames 11185–90651 and 110808–190197, from `DATA.md`) to decide where to write `X`.

3. Training data and rule compliance

- No labelled dataset of any kind was used. The only fitted models are the two logistic models of §1.4; their targets are derived from the provided *automatic* tracking, never from the possession ground truth, the video, or hand labels.
- The public ground truth was **read to understand the definition** of possession and dead ball (what a dead stretch looks like, that restarts follow stoppages), as the rules allow. It was not used to fit, tune or select any parameter. All constants are in `Params` (`sd301/pipeline.py`) and were fixed from football reasoning before looking at any score: distances in metres, durations in seconds, e.g. a hand-over needs ~1.5 s of control.
- Honest log of what the public window influenced: (a) the first run scored 0.00 spellF1 because of a sign bug in the decoder (dead and in-play costs inverted); the fix made no parameter change; (b) later changes (player-only models, restart prior, Smart Tagging) were designed from the observations in the data documentation and checked for stability with ground-truth-free tools (`tools/sensitivity.py`, `tools/diagnostics.py`), not chosen by public score. The public window has 17 possessions and cannot rank versions anyway.
- The ±3 s feature window (v4) moved the public-window spellF1 from 0.533 to 0.477 (one possession in 17: inside the noise) while leaving the full-match Smart-Tagging agreement unchanged at 0.739. We kept it because the rule we set beforehand (held-out pseudo-label AUC) favoured it, and report the drop rather than reverse the decision on 17 possessions.
- We did not compare the output against the video.

4. Possession rules implemented

- Three states only: H, A, D; D is a stoppage, never a possession.
- **Controlled possession** is approximated by *sustained* proximity: a side takes over only after about 1.5 s of net control, so touches, deflections and aerial duels do not split a possession. A loose ball stays with the last controller (state persists).
- **Restarts**: stoppage → other side most likely; set-piece Smart Tagging overrides softly.

- **Dead ball**: out of play, stationary for seconds, or players in a stoppage pattern when the ball is untracked. We do not infer dead-ball ownership.

5. Limitations

- Dead ball is over-predicted when the tracker loses the ball for long stretches without other cues (the dead share is ≈ 48 % of graded frames; expected order of magnitude for effective playing time is 55–60 min in 90 min, i.e. ~35–45 % dead). The player-only model reduces but does not remove this.
- When the ball is visible but the stoppage has begun (tracker follows a ball boy or a second ball) we call it alive.
- Independent check, ground-truth-free: the fraction of alive frames inside Smart Tagging phases that name the side we label as holder is 0.739 for this pipeline against 0.646 for `baseline_smooth50` and 0.619 for `baseline_raw` (`tools/tag_agreement.py`; the same ordering as the organisers' published full-match scores). It is a consistency indicator, not a score: the tags are automatic and phase tags are also (weak) inputs.
- Sanity of the long stoppages: the longest dead stretches we output (74-216 s) are at players' mean speed 0.8 m/s with the ball untracked, around the four goals of the match.
- Smart Tagging team tags are automatic and sometimes wrong; their weight is deliberately small.
- No video: aerial duels and tackles that need appearance cues are invisible to us.
- Next steps with more time: use the homography with the video to see the ball when the tracker cannot; learn the dead-ball and possession models on an *external* tracked match.

6. Reproduction

```
pip install -r requirements.txt            # numpy, pandas
# put the data bundle in data/ (see DATA.md); data/ is git-ignored and never published
python run_sd301.py                         # -> outputs/SD-301.csv (≈ 15 s)
python evaluation/validate_submission.py outputs/SD-301.csv
python tools/diagnostics.py outputs/SD-301.csv
python tools/sensitivity.py                 # ground-truth-free robustness table
```

Deterministic: no random numbers anywhere. Python 3.8–3.12, numpy ≥ 1.21.

7. Robustness (ground-truth-free)

Changing each hand-set constant by ±30 % keeps ≥ 97 % of graded frames unchanged and moves the possession count (277 in the reference run) between 240 and 327 (the hand-over time, `switch_s`, being by far the most influential constant). See `outputs/sensitivity.md`.